SEMINAR PRESENTATION

Artava Kshaya (Oligohypomenorrhoea)

Aetiology, diagnosis and management of scanty menstruation

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Venue
Department of Prasuti Tantra & Stree Roga seminar series

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> *Scanty periods are the most under-reported complaint in women's health and frequently the first sign of a fertility problem.*

Abstract

Artava Kshaya — scanty menstruation — is among the most under-reported complaints in Stree Roga, yet it is frequently the earliest signal of a fertility concern and is often dismissed by patients and clinicians alike. This seminar covers the classical definition and classification of Artava Kshaya, the doshic aetiology with particular reference to Vata and the role of Ama, the differential diagnosis from physiological variation and from other menstrual disorders including amenorrhoea, the investigation approach, and the therapeutic management combining ahara-vihara, aushadhi and specialised procedures such as Uttar Basti. The presentation also addresses the long-term consequences of untreated Artava Kshaya on reproductive health.

Artava Kshaya · Oligohypomenorrhoea · Hypomenorrhoea · Amenorrhoea · Prasuti Tantra · Ayurveda · Menstrual disorders · Fertility

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Topic of Presentation

ARTAVA KSHAYA/ OLIGOHYPOMENORRHEA

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ARTAVA:

Definition:

ऋतौ भवमार्तवंस्त्रीणां यदपत्यमार्गात् शुद्ध ईषत् कृष्णं विगन्धं च ।

वायु प्रेरितम् लोहितं प्रवर्तते तदार्तवमुच्यते || (अ.हृ.शा. 1/1)

The substance which flows out of the apatyamarga having red color with mild black discoloration and without any gandha due to the action of vayu in a cyclic manner or at a specific period is called “artava”.

Formation of Raja / Artava:

रसादेव स्त्रिया रक्तं रजः संज्ञा प्रवर्तते || (सु. सू. 14/6)

तथा रक्तमेव च स्त्रीणां मासे मासे गर्भकोष्ठ अनुप्राप्य त्र्यहं प्रवर्तमानं आर्तवमित्याहुः || (अ.सं.शा.1/9)

Acc to Ayurveda, the rajas or artava is formed as an upadhathu from rasa dhatu i.e., artava is formed from sara bhaga of rasa dhatu. This rakta or raja reaching garbhashaya and flows out of it periodically for 3 days is considered the artava.

ARATAVA KSHAYA

The term artava kshaya is combination of 2 words ‘artava’ + ‘kshaya’.

The word artava denotes that which occurs cyclically and kshaya from kshi dhatu which means to leave or to get reduced.

According to Ayurvediya Maha Shabdakosha,

क्षीणार्तवं नाम आर्तवस्य स्वप्रमाण हानि ||

That means the reduced quantity of artava from its normal pramana.

As a symptom artavakshaya could be observed in following yonivyapad

1. Arajaska

2. Vataja yonivyapad

Artavakshaya is defined as,

According to Acharya Sushruta,

आर्तवक्षये यथोचितकालादर्शनमल्पता वा योनिवेदना च । (Su.su.15/12)

In the event of deficiency or loss of Artava, the menstruation does not appear in it’s appropriate time or is delayed (Intermenstrual period is prolonged), is scanty, yoni vedana (abdominal pain) and does not last for 3 days.

Acharya Bhela described that even though the blood circulates throughout the body, it cannot nourish the artava due to vikrithi and alpata of rakta dhatu which in turn leads to artava kshaya.

आर्त्तवस्य स्वकाले चाभावस्तस्याल्पताऽथ वा॥

जायन्ते वेदना योनौ लिङ्ग स्यादार्त्तवक्षये ॥ (भा.प्र.पू. 7/90-91)

Bhavamishra mentions that in artavakshaya the menstrual flow either fails to occur or is scantly associated with pain in pelvic region.

Nidana

The causative factors explained in classics can be divided into for the sake of convenience, viz. Samanya (general) and Vishesha (Specific).

Samanya Nidana

In classics, Artava is considered as Updhatu of Rasa Dhatu. The factors responsible for Rasa Dhatu Kshaya are responsible for Artava Kshaya. According to Sushruta Rasa Kshaya is one of the causative factors of Dhatukshaya.

Sushruta described Mithya Ahara Vihara as causative factor for Dhatukshaya. It is Samanya Nidana of Artava Kshaya.

Vishesha Nidana

The factors responsible for Artava Utpatti if vitiated by any cause lead to Artava Kshaya. These factors are taken as Vishesha Nidana of Artava Kshaya. The Vitiated Artavotpadaka Ahara, Agni (Jathragni, Rasagni), Rasadhatu and Artavavaha Srotas are main factors for Artava vitiation. Artava is Pittapradhana Agneya in nature. So opposite Vata Kaphavardhaka Ahara Vihara are Vishesha Nidana of Artava Kshaya.

Sushruta has clearly mentioned that, main causative factor that come into action in the production of Artava Kshaya are Vata and Kapha Dosha in combined form.

Nidana of Artavakshaya

Aharaja

Viharaja

Manasika

Anashana

Ati samshodana

Shoka

Asatmya ahara sevana

Ati samshamana

Bhaya

Teekshna ahara sevana

Ati vyavaya

Chinta

Abhojana

Atapa sevana

Krodha

Atibhojana

Vega dharana

Mana santapa

Atyalpa Bhojana

Ratri Jagarana

Irshya

Divaswapna, Alasya

Purva Rupa:

कट्वम्ल लवणोष्णानि विदाहीनि गुरुणि च। फलशाकानि पानानि स्त्री काङ्क्षत्यार्त्तवक्षये ॥

(भा.प्र.पू. 7/111)

Purva Rupa of Artava Kshaya is not described in classics. In Bhavprakasha specific desire of the women suffering from Artava Kshaya have been enlisted i.e. she desire Katu, Amla, Lavana, Ushna, Vidahi, Guru Ahara and Phala Shaka and beverages etc.

Katu etc. Ahara increase Artava thus, the desire to consume these substances is probably due to deficiency of Artava. Hence it can be considered as Purvarupa of Artava Kshaya.

Rupa:

Sushruta explained the symptoms of Artava Kshaya as

Yathochita Kaala Adarshanam -

Delayed menarche- Occurring at the age of 15 years or later can be considered as Artava Kshaya.

Prolonged intermenstrual period.

Duration of menstruation less than normal duration i.e. less than 3 days.

Alpartavam –

According to Shabdakalpadruma “Alpa” means “Kshudra Pramana.” If menstrual blood flow is in less quantity i.e. less than 4 Anjali or less than 35 ml it should be considered as Artava Kshaya.

Yoni Vedana -

Due to Artava Kshaya, vitiation of Vata mainly Apana and Vyana Vata causes Yoni Vedana. Yoni Vedana can be considered as spasmodic, radiating, and infrequent pain during menstruation.

Samprapti:

2140585151733

NIDANA SEVANA 0

NIDANA SEVANA

919354974500038320231149270 0

297576463122

-19848183226

Kapha Prakopaka Nidana 00

Kapha Prakopaka Nidana 2144764163468

Vata Prakopaka Nidana sevana 0

Vata Prakopaka Nidana sevana

43289324405

Vata-Kapha Prakopaka Nidana 0

Vata-Kapha Prakopaka Nidana

29857701506965198745155594766405468l4

left95059

Agnimandya (deranged metabolism) 0

Agnimandya (deranged metabolism)

214503025193

Vata Prakopa (mainly Apana Vayu) 0

Vata Prakopa (mainly Apana Vayu) 434530575584

Vitiation of VataKapha 0

Vitiation of VataKapha

300764916666552152551319557564022815 2

1587577345

Ama Utpatti 0

Ama Utpatti

216344565367

Apana Vayu vitiated in Yoni Pradesh 0

Apana Vayu vitiated in Yoni Pradesh 438023045431

Avarana of Apana Vayu by Vitiated Kapha 0

Avarana of Apana Vayu by Vitiated Kapha

7573536218

left65702

Disturbance in formation of Rasa Dhatu 00

Disturbance in formation of Rasa Dhatu

29883101174945229225785010 0

4417060159322

Artava-vaha Srotas obstruction (Sanga/ Avarodha) by vitiated Kapha 0

Artava-vaha Srotas obstruction (Sanga/ Avarodha) by vitiated Kapha

757555164471216377870655

Aartava Pravartan Karma of Apana Vayu is hampered 0

Aartava Pravartan Karma of Apana Vayu is hampered

-1778041162

Rasa Dhatu Kshaya 0

Rasa Dhatu Kshaya

782175567733022254103039005266055214964 00

-8539864443

Rasa-Agni Vaishmya 0

Rasa-Agni Vaishmya 2209347154632

Artava Adarshan/ Alpata, Yoni vedana 0

Artava Adarshan/ Alpata, Yoni vedana 4420870255371

ARTAVA KSHAYA 0

ARTAVA KSHAYA

766445194417

-27160238709

Updhatu (Artava) formation decreased 00

Updhatu (Artava) formation decreased

784986234950

-27940286863

Artava Kshaya (Alpata/ Kshudra) Pramana) 00

Artava Kshaya (Alpata/ Kshudra) Pramana)

Samprapti ghataka:

Dosha

Avarita –Vata (Apana and Vyana), Avaraka –Kapha

Dushya

Rasa, Rakta

Upadhatu

Artava

Agni

Jathragnimandhya; Dhatvagnimandhya

Srotas

Rasavaha, Artavavaha

Srotodusti

Sanga

Adhisthana

Garbhashaya, Beejagranthi

Rogamarga

Abhyantara

Roga Prakruti

Chirakari

Sapeksha Nidana:

Artava kshaya explained as a symptom of many yonivyapad. As it is essential to confirm the diagnosis beforehand of treatment, the differentiation of condition with similar pathology is done as below.

Vataj yonivyapad: In this condition, lakshanas like ayama, suptaa, pipilika sarana, yonikarkashata are seen which are not concerned for artavakshaya.

Arajaska yonivyapad: Symptoms like daha, karshya, vaivarnya are in Arajaska.

Vandhya: It is nashtartava but not ksheenartava.

Shandi: Stana abhava is factor for differentiation.

Lohita kshaya: In this condition, ksheena rajas with daha etc are profoundly observed.

Sadhya-Asadhyata

In Ayurvedic classics, prognosis of Artava Kshaya is not mentioned but Sushruta explained treatment of Artava Kshaya.

By which it can be interpreted that Artava Kshaya is either Sadhya or Kashta-Sadhya Vyadhi.

Chikitsa

Chikitsa is Samprapti Vighatana. In Artava Kshaya, there is vitiation or Vruddhi of Vata and Kapha along with Kshaya of Pitta, Rasa and Rakta Dhatu due to Agnimandya. Hence all these Samprapti Ghataka should be considered to treat Artava Kshaya.

Nidana Parivarjana is the basic management of all diseases. In Artava Kshaya, Hetus should be avoided in order to get permanent relief.

तत्र संशोधनमाग्नेयानां च द्रव्याणां विधिवदुपयोगः। (सु.सं.सू. 15/12)

' संशोधनमिह सामान्यम्' इति केचित्, 'संशोधनमिह वमनं न विरेकः इत्यपरे', कुताः? विरेचनेन हि पित्तक्षयादार्तवस्य क्षय एव स्यादिति, वमनेन तु सौम्यधातौ निर्हते आग्नेयधातौ वृद्धे आर्तवमाप्याय्यते। आग्नेयानां च द्रव्याणामिति तिलमाषसुराशुक्तादीनाम्। (सु.सं.सू. 15/12, डल्ह.टी.)

Artava Kshaya should be treated with Samshodhana with Agneya Aushadhi. Dalhana explained Shodhana should be done by Vamana and not by Virechana. Because Virechana reduces Pitta that reduces Artava. Vamana eliminate Saumya Dhatu-Kapha resulting into a relative increase in Agneya Dhatu of body, consequently Artava quantity increases.

तत्र शोधनमित्यनेन स्त्रोतःशुद्धयर्थं शोधनं, तच्च वमनं विरेचनं चोर्ध्वाधः स्त्रोतः शुद्धिकरताय..... विधिवदित्यनेन शोधनपूर्वकमुपयोगं मात्रादिना योग्येन युक्तं दर्शयति ।। (सु.सं.सू. 15/12, चक्र.टी.)

As per Chakrapani Tika - Purifying measures clear Srotasa (Channels). Vamana Karma and Virechana Karma clears upward and downward directed channels respectively, thus both should be used, giving due consideration to the dosages of drugs used for purification and fitness of the patient.

क्षीणं प्रागीरितं रक्तं सलक्षणचिकित्सितम्। तथाऽप्यत्र विधातव्यं विधानं नष्टरक्तवत्॥

(सु.सं.शा. 2/23, डल्ह.टी.)

Aacharya Sushruta clarifies that kshinartava should be treated like nashtartava.

अल्पपुष्पा (अनुवास्या) इति। (का.सं.सि. 7/11)

Aacharya Kashyapa says artava dushti is an anuvasana yogya vyadhi.

In Artava Kshaya there is qualitative Kshaya of Pitta and quantitative Kshaya of Artava. The use of Agneya Dravyas is beneficial in Artava Kshaya because of Agnivardhaka and Pittakara properties that helps in Samprapti Vighatana.

Artavanasha is a basti-sadhya roga.

तत्र मत्स्यकुलत्थाम्लतिलमाषसुरा हिताः। पाने मूत्रमुदश्वित्च्च दधि शुक्तां च भोजने ॥

(उदश्वित् अर्धोदकं तक्रम्। शुक्तं चुक्रम्-डल्हणः) ॥ (सु.सं.शा. 2/22, डल्ह.टी.)

Food products advised are fish, kulattha, kanji, tila, masha, gomutra, udashvita (butter milk with half water), curd and sukta.

Lashuna, shatavari, shatapuṣpa are beneficial in all the artavavyapad.

Varti prepared from ikshvaku bija, danti, chapala, guda, madanaphala, kinva, yavashuka pestled with snuhikshira enables expulsion of artava.

Jyotishmati patra, swarjikakshara (rajika), ugra and asana with cold water for 3 days

Kwatha of krushna tila with guda taken daily morning induces menstruation in chronic cases.

Kwatha of tila, shelu, karavi with guda induces menses within 3 days.

Phala ghrita and brihat shatavari ghrita is beneficial.

Shatavaryadi anuvasana basti is beneficial.

According to Vngasena Samhita, (व.सं., स्त्रीरोगाधिकार 73/1-4)

Wine mixed with kalka of maricha, asana (sprouts) is beneficial.

Yonivarti prepared from paste of iksuvaku seeds, danti bija, pippali, guda, madanaphala, yeast, yavakshara pounded with snuhi kshira is beneficial.

Kalka of japapuspa or jyotishmati leaves pounded with sour gruel when administered generates menstrual flow.

Kalka made from roots of lotus, yeast pounded together and taken with cold water.

लाङ्गलीकन्दचूर्ण वा मूलं वाऽप्यपमार्गजम्।

इन्द्रवारुणीमूलं वा योनिस्थं पुष्परोधनुत् ॥ (२.र.स. 25/87)

Yonipurana with langali kanda churna or apamarga mula churna or indravaruni mula churna induces menstruation.

तिलमूलकषायं तु ब्रह्मदण्ढीयमूलकम्। यष्टीत्रिकटुकं चूर्ण क्काथयुक्तं हि पाययेत्।।

पुष्परोधे रक्तगुल्मे स्त्रीणां सधः प्रशस्यते । (र.र.स. 25/88)

Tila mula kashaya (1/2 pala) and brahma dandhiya mula kashaya (1/2 pala) or 3 masha churna along with yashti and trikatu (3 masha each) when administered ensures menstruation in pusparodha and raktagulma.

चणकानां रसं चैव शृतं क्षीरं घृतेन वा। शर्करामधुसंयुक्तं रक्तदोषहरं पिबेत् ।। (र.र.स. 25/89)

Intake of chanaka swarasa or kwatha with milk or ghee or sugar or honey or altogether cures all types of rakta dosas (especially raktagulma etc.)

तिलक्काथा गुड व्योषं तिलभाङ्गींयुतं पिबेत्। क्कार्थ रक्तभवे गुल्मे नष्टपुष्पेऽथ पाययेत् ।। (र.र.स. 25/90)

तिल क्काथो घृतगुडव्योष भार्डी रजोन्वितः। पानं रक्तभवे गुल्मे नष्टेपुष्पे च योषितः ॥

(वृ.मा. 30/57, रा.मा. 31/8)

Purana guda, trikaṭu churṇa, tila churṇa, bharangi churna are taken in equal quantity made into bolus measuring 2 tola each is taken daily morning followed by tila mula kwatha; it is beneficial in raktagulma and nashtartava.

इक्ष्वाकुबीजदन्तीचपलागुडमदनफलकिण्वयष्ट्याह्वम्।

सस्नुक्क्षीरैर्वर्त्तिर्योनिगता कुसुमसञ्जननी।। (भै.र. 67/27)

Yoni varti prepared from ikshvaku bija, danti, pippali, guda, madanaphala, kinva (surabija), madhuyashti churna with snuhi kshira; induces menstruation.

सकाञ्जिकं जपापुष्पं भृष्टं ज्योतिष्मतीदलम्। दूर्वापिष्टञ्च सम्प्राश्य वनिता त्वार्त्तवं लभेत्॥

(भै.र. 67/28, च.द. 62/22)

Kalka prepared from japa pushpa, leaves of jyotishmati and fried in ghrita, then titurated in durva swarasa. It is effective in inducing menstrual flow.

पीतं ज्योतिष्मतीपुष्पस्वर्जिकोग्रासनं त्र्यहम्। शीतेन पयसा पिष्टं कुसुमं जनयेद् ध्रुवम् ।। (भै.र. 67/29)

Jyotishmati pushpa, svarjikshara, vacha, shala puspa made into paste with water and administered for three consecutive days induces menstrual flow.

द्वयक्षाणि वेणु पर्वाणि शुकपुष्पा पलोन्मिता। गुडेन मधुरः क्काथः प्रवर्तयति वैरजः ।। (सि.भे. 4/1105)

Oral intake of kwatha made from bamboo nodes (20g) and shatapuspa (40g) mixed with guda induces menstruation.

HYPOMENORRHEA

Hypomenorrhea also known as short or scanty periods, is extremely scanty or less menstrual blood flow. Uterine bleeding is slight or less in amount or duration or both. Bleeding that lasts for less than 2 days is termed as Hypomenorrhea.

CAUSES:

Constitutional:

Many times, scanty menses is found in the whole menstrual cycle and is of no clinical significance. It is perhaps due to presence of unusual insensitive arrangement of endometrial vascular apparatus.

Uterine:

It may follow myomectomy, intrauterine adhesions, or any other operations on the uterus where endometrial cavity is reduced. Endometrial tuberculosis forms 50% case of hypomenorrhea in women, hence it is vital to perform preliminary hysteroscopic examination in hypomenorrhic before attempting IVF. A positive case of Tb is managed by lysis of adhesions followed by IUCD insertion.

Hormonal:

Hypomenorrhea may precede amenorrhea. It may also happen in cases with long term use of low dose OCP leading to progress endometrial atrophy.

Nervous and emotional:

Psychological factors failing to supress ovarian and uterine cycle completely succeed only in reducing the amount of flow as seen in pseudocyesis where hypomenorrhea is more common than amenorrhea.

PATHOPHYSIOLOGY:

295980492980004996192109220141832488929Pituitary dysfunction Unresponsive FSH Ovarian unresponsiveness hypomenorrhea.

Undue prolongation of proliferation phase with normal secretory phase.

Criteria For Diagnosis:

Menstrual bleeding occurring more than 35 days apart

Remains at the same frequency

Unduly scanty

Menstrual flow lasting for less than 2 days.

Examination:

General examination, abdominal examination and pelvic examination is mandatory in all women having hypomenorrhea.

Investigations:

Urine pregnancy test to exclude pregnancy

General routine check-up blood, urine and stool to rule out any infection.

Per speculum and per vaginal examination.

Hormonal assessment for FSH, LH, serum PRL, testosterone, thyroid levels

Ultrasonography

C.T scan

Laparoscopy

Chromosomal study

TREATMENT:

Mainly depends on the cause

Reassurance and psychotherapy are very useful in psychogenic and environmental cause.

Anaemia should be treated with appropriate medications

In case of malnutrition – management with proper diet

In case of obesity – weight loss therapy, treating the cause

Systemic illness should be managed adequately

Hormonal therapy & surgical therapy.

OLIGOMENORRHOEA

Definition

Oligomenorrhoea can only be defined arbitrarily as one in which the cycle lasts longer than 35 days. Menstruation may be both infrequent and irregular, or may be regularly infrequent.

In either case the error is in the ovary and its controlling factors, rather than in the uterus; for this reason, the symptom tends to have more significance than does hypomenorrhoea. Infrequent menstruation and amenorrhoea are essentially similar symptoms with identical causes—the difference is only one of degree. Indeed, when menstruation only occurs at intervals of 3 or more months, it is a matter of opinion as to whether the condition is described as one or the other.

Causes

Constitutional and Physiological

In the majority of cases infrequent menstruation represents a peculiarity of the individual and is not out of keeping with health and good fertility. It can be familial. The bleeding can be ovular in type, which means that the ovarian cycle is drawn out or temporarily arrested at some phase. As a rule, the luteal phase tends to be fairly constant at 14 days; it is the follicular phase which is either lengthened or slow to commence. Infrequent menstruation sometimes follows the menarche and precedes the menopause; it then marks a gradual waxing and waning of the endocrine cycle, and is physiological. Adolescent oligomenorrhoea tends to cure itself within a few years but may persist until the first pregnancy, after which it is often replaced by a more normal cycle.

Hormonal

If there is any abnormality it is essentially one in which the ovary is underactive, and this is reflected in the secondary sex organs. Hypoplasia of the uterus and vagina, and a history of a late menarche, are therefore to be expected. The ovarian disturbance is often secondary to hypothalamic, pituitary, thyroid or adrenal dysfunction, so the patient may have other stigmas of endocrine upset such as obesity, squat figure, hirsutism and low fertility. Obesity and oligomenorrhoea are close companions. So are oligomenorrhoea and hirsutism, but the basis may still be genetic and familial. Otherwise these combinations may be part of the PCOS.

Chromosomal

An underlying sex chromosome abnormality, such as an XXX arrangement, is occasionally found.

Treatment

The investigation and treatment of oligomenorrhoea are the same as those for amenorrhoea, but the prognosis is better because gross abnormalities of the ovary and uterus are less likely to be present.

There is no need for any treatment except reassurance in the majority of cases, that is, those in which the condition appears to be constitutional and is unassociated with impaired fertility.

Over treatment should also be avoided, particularly in young girls in whom a spontaneous cure can be expected. If, in addition, infertility is a problem, treatment may be required to induce or increase the frequency of ovulation. Counselling regarding the time of intercourse is difficult with erratic or infrequent menstrual cycles.

FORMULATIONS: -

Rajahpravartini vati

Nashtpushpantaka Rasa

Dashamoola kwatha

Lashuna Vati

Sukumara Ghrita

Bruhat Kalyanaka Ghrita or Kalyanaka Ghrita

Kumaryasava

Kanchanar guggulu

Shatapushpa churna

CONCLUSION

Artava kshaya occurs due to involvement of vata and kapha, which can be attributed to marga avarodha of artava vaha strotas. It can significantly impact female reproductive health and is often associated with conditions like oligohypomenorrhea. In Ayurvedic classical texts, treatment is mentioned to keep the doshas in equilibrium. It can be achieved by Samsodhana (vaman and basti) and Samshamana (in form of Agneya dravyas). Early diagnosis and intervention are crucial to restore menstrual regularity and improve overall fertility outcomes. Use of Agneya dravyas not only relieves the kapha which does avarana to apanavata but also increases the quantity of artava.

REFERENCES: -

Stree roga Gynecology by Dr. Hemalatha Kapoorchand

Stree roga textbook by PV tiwari

DC Dutta’s textbook of Gynecology

Jeffcoate’s PRINCIPLES OF GYNAECOLOGY

Williams Gynecology

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